

Dashboard Implementation Documentation

Application: Lodgings and Restaurant Management

Feature: Admin analytics dashboard for stored partner API and partner service data.

Dashboard Screenshot

The dashboard is available to admin users at `/api-dashboard` . The screenshot below shows the implemented layout with representative stored data.

STORED DATA ANALYSIS

Integration Analytics

Monitor partner traffic, stored service inventory, reliability, and AI-generated operations insight.

Dashboard Logout

TOTAL CALLS
128
Requests received

SUCCESSFUL
119
2xx responses

FAILED
9
Non-2xx responses

SUCCESS RATE
93%
Overall API reliability

ACTIVE SERVICES
42
Listed to partners

CANCELLED
6
Removed from listing

LODGING
27
Active lodging services

RESTAURANT
15
Active restaurant services

GROQ LLM FEATURE
AI Operations Insight
Generate a concise admin summary from stored API usage and partner service data.

Generate Insight

24 Hour Health
Current request volume and failures.

REQUESTS
36

FAILURES
2

Latest AI Insights
Persisted Groq responses.

AI operations insight
Partner traffic is healthy with a 93% success rate. Lodging services dominate the active catalog, while two recent failures should be reviewed for partner onboarding quality.
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Partner Usage
Traffic volume by integration partner.

Zomato
96% successful
54

BookingHub
91% successful
43

DineConnect
87% successful
31

Service Mix
Active services and average partner-visible price.

LODGING
27
Average price: 4799.0

RESTAURANT
15
Average price: 899.0

Purpose

The dashboard helps administrators understand how third-party integrations are using the application and what partner-visible services are currently stored. It analyzes data already

persisted in the application's database rather than relying on hard-coded values.

Displayed Analysis

Dashboard Area	Data Source	Explanation
API overview cards	<code>api_usage_logs</code>	Total calls, successful calls, failed calls, and success rate are calculated from stored API log rows.
Stored service overview	<code>partner_services</code>	Shows active services, cancelled services, active lodging records, and active restaurant records.
24 hour health	<code>api_usage_logs.requested_at</code>	Counts recent traffic and failures during the last 24 hours.
Endpoint usage	<code>api_usage_logs.endpoint</code> , <code>method</code> , <code>successful</code>	Groups API requests by endpoint and HTTP method.
Partner usage	<code>api_usage_logs.partner_name</code>	Groups traffic by partner and renders a success-rate bar.
Service mix	<code>partner_services.service_type</code> , <code>price</code>	Groups active services by type and calculates average price.
Recent services and requests	<code>partner_services</code> , <code>api_usage_logs</code>	Shows the latest stored records for operational review.

Implementation Details

The dashboard route is implemented in `ApiDashboardController` . It queries repositories, computes success-rate percentages, and places the resulting values into the Thymeleaf model.

The repository layer was extended with aggregate methods and JPQL projection queries:

- `ApiUsageLogRepository` : total, 24-hour counts, endpoint summaries, and partner summaries.
- `PartnerServiceRepository` : active/cancelled counts, service type counts, recent services, and active service type price summary.
- `EndpointUsageSummary` , `PartnerUsageSummary` , and `ServiceTypeSummary` : typed projection records for dashboard data.

The UI is implemented in `src/main/resources/templates/api-dashboard.html` and styled through `src/main/resources/static/css/auth.css` . The page uses responsive grids, summary cards, tables, and bar indicators so it remains usable on desktop and mobile screens.

Access Control

The dashboard is protected in `SecurityConfig` with `.requestMatchers("/api-dashboard").hasRole("ADMIN")` . Normal users can continue to use the general dashboard, while only admins can view integration analytics.

Verification

The project test suite was run with `./gradlew.bat test` and completed successfully after implementation.